Kernix 0.6.0 / Performance brief

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

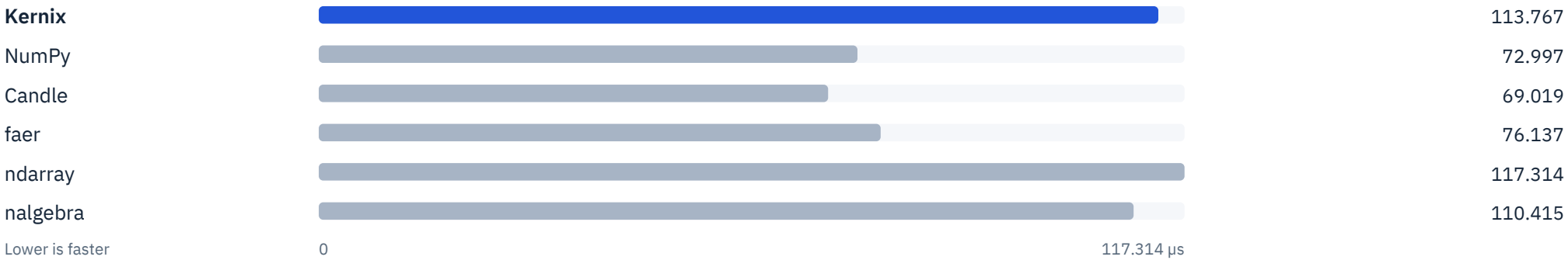
09 workloads

40 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

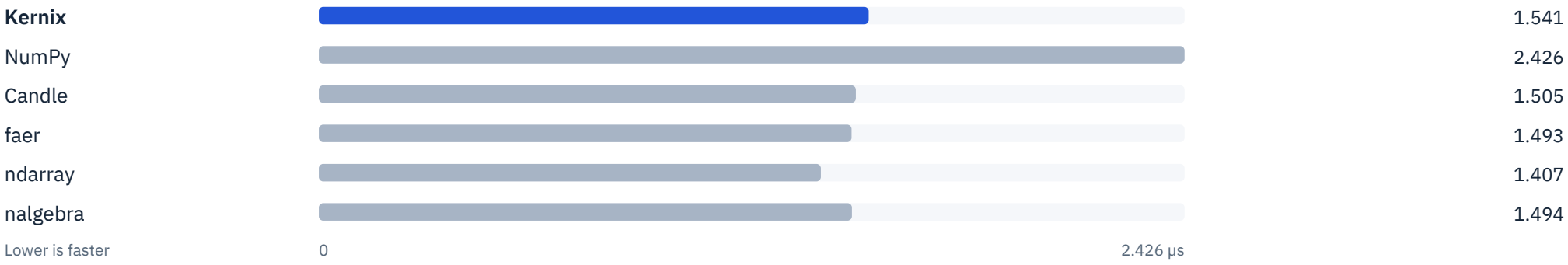
Kernix 0.6.0 / Matrix multiplication

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product

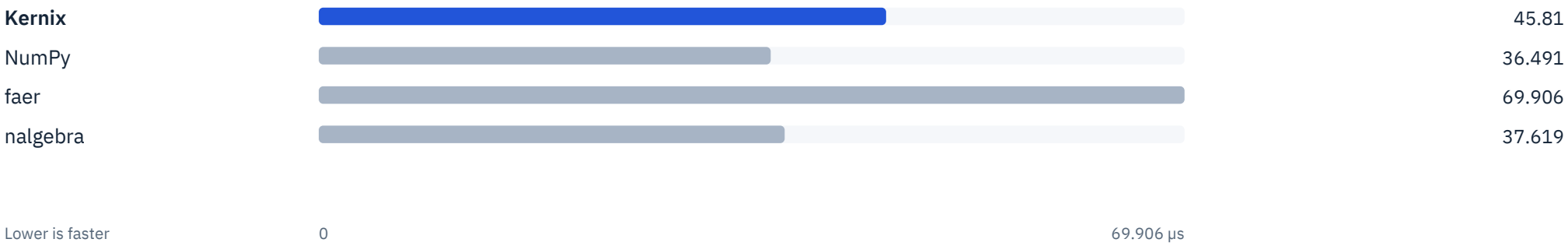


Kernix 0.6.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

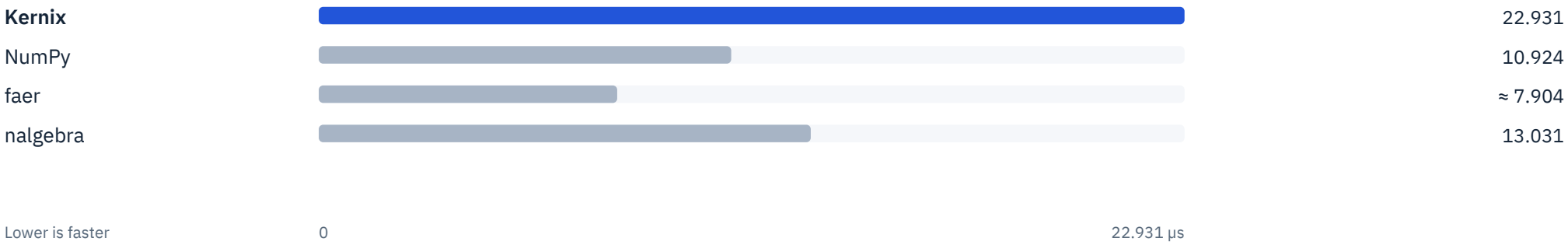
256 × 64 · f64 · 1 worker · allocated output

Kernix 0.6.0 / Decompositions & solvers

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



Kernix 0.6.0 / Training & autodiff

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

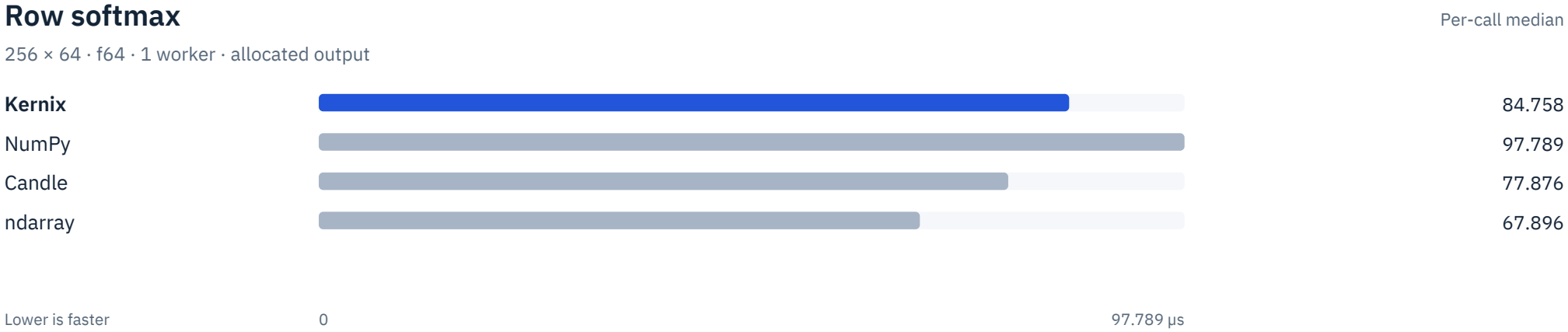
sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product

Kernix 0.6.0 / Statistics & activations

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



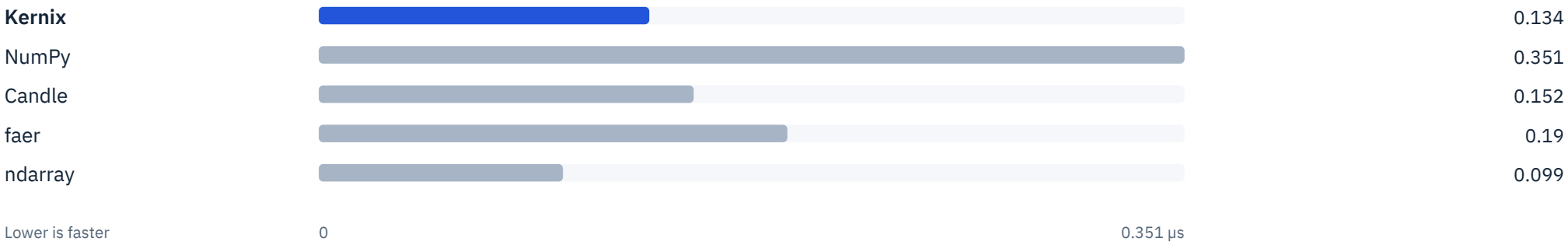
Kernix 0.6.0 / Tensor construction

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

64 × 64 · f32 · 1 worker · allocated output

Per-call median



Create ones

64 × 64 · f32 · 1 worker · allocated output

Per-call median

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	1541	1.541
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	1407	1.407
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	1494	1.494
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	1493	1.493
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	1505	1.505
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	2426	2.426
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	113767	113.767
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	117314	117.314
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	110415	110.415
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	76137	76.137

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	69019	69.019
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	72997	72.997
Thin QR / Kernix 96 × 32 · f64	45810	45.81
Thin QR / nalgebra 96 × 32 · f64	37619	37.619
Thin QR / faer 96 × 32 · f64	69906	69.906
Thin QR / NumPy 96 × 32 · f64	36491	36.491
Thin QR / Kernix 256 × 64 · f64	468986	468.986
Thin QR / nalgebra 256 × 64 · f64	356914	356.914
Thin QR / faer 256 × 64 · f64	393041	393.041
Thin QR / NumPy 256 × 64 · f64	326424	326.424

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	22931	22.931
Cholesky factorization / nalgebra 64 × 64 · f64	13031	13.031
Cholesky factorization / faer 64 × 64 · f64	7903.999999999999	7.903999999999999
Cholesky factorization / NumPy 64 × 64 · f64	10924	10.924
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3165	3.165
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	3914	3.914
Row softmax / Kernix 256 × 64 · f64	84758	84.758
Row softmax / ndarray 256 × 64 · f64	67896	67.896
Row softmax / Candle 256 × 64 · f64	77876	77.876
Row softmax / NumPy 256 × 64 · f64	97789	97.789

Recorded precision / exact latency table

12th Gen Intel(R) Core(TM) i3-1215U · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / Kernix 64 × 64 · f32	134	0.134
Create zeros / ndarray 64 × 64 · f32	99	0.099
Create zeros / faer 64 × 64 · f32	190	0.19
Create zeros / Candle 64 × 64 · f32	152	0.152
Create zeros / NumPy 64 × 64 · f32	351	0.351
Create ones / Kernix 64 × 64 · f32	183	0.183
Create ones / ndarray 64 × 64 · f32	155	0.155
Create ones / faer 64 × 64 · f32	171	0.171
Create ones / Candle 64 × 64 · f32	238	0.238
Create ones / NumPy 64 × 64 · f32	1221	1.221